

Sentinel Energy Partners

Monthly Production & Compliance Intelligence

Report Type	Monthly Energy Production & Price Risk Report
Coverage Period	2026-03 (latest EIA data)
Data Sources	US Energy Information Administration (EIA) + Yahoo Finance
Coverage	US Crude Oil (MBBL/D) + Natural Gas (MMCF) + WTI & Henry Hub futures
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Generated	2026-06-03 00:12:27

SAMPLE REPORT — LIVE DATA

1. Executive Summary

- **Crude Production (Latest):** 13,696.0 MBBL/D (2026-03) — MoM: -0.01% | YoY: 0.00%
- **Natural Gas (Latest):** 3,789,520 MMCF (2026-03) — MoM: +11.29% | YoY: +7.47%
- **WTI Crude Oil:** \$94.72/bbl — 6mo range \$55.27–\$112.95 | Change: +60.68%
- **Henry Hub Nat. Gas:** \$3.169/MMBtu — 6mo range \$2.523–\$7.460 | Change: -36.56%
- **Revenue at Risk (95%):** Crude \$12,013.9M | Natural Gas \$6,811.2M — based on current production and price volatility

DATA AUDIT TRAIL — Generated: 2026-06-03 00:12:27
Crude: EIA API v2 — api.eia.gov/v2/petroleum/crd/crpdn/data/ (MBBL/D, NUS, EPC0)
Natural Gas: EIA API v2 — api.eia.gov/v2/natural-gas/prod/sum/data/ (VGM, NUS, MMCF)
WTI: Yahoo Finance (CL=F) via [yfinance](#) | Henry Hub: Yahoo Finance (NG=F) via [yfinance](#)
All data publicly available and independently verifiable.

2. US Crude Oil Production

13,696.0

LATEST (MBBL/D)

-0.01%

MONTH-OVER-MONTH

0.00%

YEAR-OVER-YEAR

Trailing 6 Months

Period	Production (MBBL/D)	MoM Change	Units
2026-03	13,696.0	—	MBBL/D
2026-02	13,697.0	+0.01%	MBBL/D
2026-01	13,305.0	-2.86%	MBBL/D
2025-12	13,657.0	+2.65%	MBBL/D
2025-11	13,789.0	+0.97%	MBBL/D
2025-10	13,864.0	+0.54%	MBBL/D

Source: EIA API v2 — api.eia.gov/v2/petroleum/crd/crpdn/data/ | Pulled: 2026-06-03 00:12:27

3. US Natural Gas Production

3,789,520
LATEST (MMCF)

+11.29%
MONTH-OVER-MONTH

+7.47%
YEAR-OVER-YEAR

Trailing 6 Months

Period	Production (MMCF)	MoM Change
2026-03	3,789,520	—
2026-02	3,405,144	-10.14%
2026-01	3,693,462	+8.47%
2025-12	3,800,813	+2.91%
2025-11	3,648,204	-4.02%
2025-10	3,675,081	+0.74%

Source: EIA API v2 — api.eia.gov/v2/natural-gas/prod/sum/data/ | Pulled: 2026-06-03 00:12:27

4. Commodity Pricing Dashboard

Commodity	Ticker	Current	6mo High	6mo Low	6mo Change	Ann. Vol	VaR 95% (1mo)
WTI Crude Oil	CL=F	\$94.72/bbl	\$112.95	\$55.27	+60.68%	65.0%	-28.72%
Henry Hub Nat. Gas	NG=F	\$3.169/MMBtu	\$7.460	\$2.523	-36.56%	119.4%	-56.92%

Source: Yahoo Finance (yfinance) — finance.yahoo.com | CL=F (WTI front-month futures), NG=F (Henry Hub front-month futures), 6-month daily OHLCV | Pulled: 2026-06-03 00:12:27

5. Revenue at Risk Analysis

Commodity price volatility translates directly into revenue variance for E&P operators. The Revenue at Risk (RaR) metric quantifies the expected maximum monthly revenue shortfall at a given confidence level, based on current production volumes and price volatility.

Revenue at Risk — 95% Confidence (Monthly)

Commodity	Production	Price	Est. Monthly Revenue	Price Vol (Ann.)	Monthly Vol	Revenue at Risk (95%)
WTI Crude Oil	13,696.0 MMBL/D	\$94.72/bbl	\$38,918.6M	65.0%	18.8%	\$12,013.9M
Natural Gas	3,789,520 MMCF/mo	\$3.169/MMBtu	\$12,009.0M	119.4%	34.5%	\$6,811.2M

Crude oil interpretation: At 95% confidence, monthly crude oil revenue at risk is approximately \$12,013.9M based on 65.0% WTI annualized volatility and 13,696.0 MMBL/D national production. This represents the expected tail loss in a 1-in-20 adverse price scenario over a calendar month.

Natural gas interpretation: At 95% confidence, monthly natural gas revenue at risk is approximately \$6,811.2M based on 119.4% Henry Hub annualized volatility. Natural gas historically exhibits higher price volatility than crude oil, making hedging programs particularly valuable for gas-weighted producers.

Revenue at Risk methodology: $\text{Monthly Revenue} \times 1.645 \times (\text{Annualized Vol} / \sqrt{12})$. Production source: EIA API. Price source: Yahoo Finance (yfinance). VaR is parametric at 95% confidence level. Actual operator revenue depends on specific production mix, contract terms, and regional basis differentials.

6. Risk Assessment

Risk Factor	Level	Commentary
WTI Price Volatility	HIGH	Annualized WTI vol 65.0% — elevated hedging cost environment
Nat. Gas Price Risk	HIGH	Henry Hub vol 119.4% — gas historically more volatile than crude
Production Trend	WATCH	US crude declining YoY (0.00%) — monitor for supply tightening
Revenue at Risk (Crude)	HIGH	95% VaR: \$12,013.9M/month at current production and WTI volatility
Revenue at Risk (Gas)	MEDIUM	95% VaR: \$6,811.2M/month — high NG volatility warrants fixed-price hedging

7. Recommendations

- Price environment:** WTI at \$94.72/bbl — strong netback; optimize production timing and lock in hedges
- Hedging priority:** With \$12,013.9M monthly crude Revenue at Risk at 95% confidence, producers >500 MMBL/D should consider collar or fixed-price contracts for Q3/Q4

3. **Natural gas:** Henry Hub at \$3.169/MMBtu with 119.4% vol — consider locking hedges for seasonal demand
4. **Production monitoring:** MoM crude change -0.01% — no anomalies
5. **Compliance:** Maintain monthly EIA production data submissions per 30 CFR 1202

8. Methodology & Data Sources

Component	Method	Source
Crude Production	EIA API v2 — petroleum/crd/crpdn, duoarea=NUS, product=EPC0, units=MBBL/D	api.eia.gov
Natural Gas Production	EIA API v2 — natural-gas/prod/sum, duoarea=NUS, process=VGM	api.eia.gov
WTI Crude Price	Yahoo Finance — CL=F (front-month WTI futures), 6-month daily via yfinance	finance.yahoo.com
Henry Hub Price	Yahoo Finance — NG=F (front-month Henry Hub futures), 6-month daily	finance.yahoo.com
Volatility	$\sigma(\text{daily returns}) \times \sqrt{252}$ — annualized daily return standard deviation	Computed
Monthly VaR (95%)	Parametric: $(\text{mean_daily} - 1.645 \times \text{daily_vol}) \times \sqrt{21}$	Computed
Revenue at Risk	Monthly Revenue $\times 1.645 \times (\text{Ann. Vol} / \sqrt{12})$ — parametric 95% confidence	Computed
MoM/YoY Changes	% change vs. prior month and same month prior year (EIA series)	Computed
Report Period	Latest EIA data (2026-03) + 6-month commodity price window	—

All data publicly available. EIA: eia.gov | Yahoo Finance: finance.yahoo.com | Pulled: 2026-06-03 00:12:27

Disclaimer & Disclosure: This report uses **live production and pricing data from the US Energy Information Administration (EIA) and Yahoo Finance** at generation time. "Sentinel Energy Partners" is a fictional firm name used for demonstration — all underlying data is real and independently verifiable. Revenue at Risk figures are statistical estimates based on national production data and exchange futures prices; actual operator results depend on production mix, contract terms, and basis differentials. Does not constitute investment advice or production forecast.

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